

Varun Khurana

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 London, UK

Meticulous Data Professional (MSc Data Science) with 2 years of experience in data management, predictive modelling, and process optimisation. With a proven expertise in handling large-scale datasets, ensuring data integrity, and automating pipelines using Python, SQL, and Azure.

Education

Newcastle University | *Master of Science in Data Science and AI* | *Vice Chancellor Scholarship Recipient* **December 2025**

- Coursework Focus: Big Data Analytics, Probability & Statistics, Database Systems, Statistical Learning, Engineering for AI, Deep Learning & Advanced AI
- Achievement: Winner *KCSxNUCATS* Hackathon held at Newcastle University (Please Check: Warehouse Reallocation Optimisation)

Manipal University Jaipur | *Bachelor of Technology in Electronics and Communications Engineering* **August 2023**

- Coursework Focus: Embedded Systems & RTOS, Digital Signal Processing (DSP), Control Systems, Internet of Things (IoT) & Sensor Circuits, and Microcontrollers.

Skills

Machine Learning & AI: PyTorch, Scikit-learn, Transformers, LLMs, Predictive Modelling

Data Management: Data Cleaning, Validation, ETL Pipelines, Data Integrity Checks, Query Management

Software & Tools: SQL, Power BI, Azure DevOps, Docker

Programming: Python (Pandas, NumPy) – Expertise in scripting for data automation

Analysis: Statistical Analysis, Anomaly Detection, Reporting & Visualisation

Experience

CodaBot (Ed Tech) **August 2023 - May 2024**

Backend Developer (AI)

Tech Stack: PyTorch, NLTK, spaCy, Azure Blob Storage, Docker, Azure DevOps, Azure Services

Hong Kong (Remote)

- Developed a Django-ReactJS app to enable real-time ML model integration, delivering predictions with **sub-1 second** latency
- Engineered ETL pipelines processing **15M+ records** daily, automating ingestion, transformation, and validation, **saving 3-man hours** per dataset
- **Delivered an Automated essay scoring (AES)** system using a fine-tuned **RoBERTa** transformer-based model on a diverse, annotated dataset, improving scoring accuracy and processing efficiency.
- Led **PoC** initiatives for **Azure-based** deployment and CI/CD pipelines (DevOps + Blob Storage), enabling scalable, monitored rollout of ML services with zero downtime.

Central Research Laboratory (Ministry of Defence, Govt. of India)

March 2023 - July 2023

Data Scientist Intern (R&D)

Tech Stack: Python, Shell, PyQt, Matplotlib, SerialPy, SQL, Docker

India

- Designed a predictive maintenance solution for automated token systems for client DMRC using LSTM-based models, reducing unplanned downtimes by **~40%** and improving overall equipment availability to **~98.5%** through automated alerts and repair workflows.
- Led a team of 3 to develop a **classification system for aerial objects** (civilian aircraft / military aircraft / missile) using time-series radar signature data; implemented ensemble ML models achieving **~88% classification accuracy**.
- Developed multiple **PoCs** for chatbot automation over WhatsApp via **Twilio**, reducing manual service desk operations and accelerating user engagement.

- . Built a Python desktop app under 30MB to decode **1.5M+ bytes/sec** from RS232 ports, enabling real-time LiDAR and GPS data analysis
- . Engineered **25+ algorithms** for rut measurement analysis using LiDAR, enhancing sensor data interpretation and visualization accuracy

Projects

Food Identification with Segmentation – *Stack: PyTorch, Albumentations, OpenCV, scikit-learn*

- . Developed a **semantic segmentation model** using a **ResNet50 encoder** paired with an **attention-guided decoder** for pixel-level classification of **104 food classes**.
- . Employed **advanced data augmentation** and **mixed-precision training**, achieving a validation **mIoU of 0.69** along with detailed per-class performance analysis.
- . Visualised **segmentation outputs** and **quantitative metrics** to reveal strengths and challenges, providing valuable insights for future improvements.
- . Proposed enhancements such as **multi-scale feature fusion** and refined **augmentation strategies** to further boost accuracy.

Multimodal Idiomaticity Representation – *Stack: PyTorch, Transformers (GPT-2, CLIP), Hugging Face, Numpy*

- . Designed a **multimodal system** that augments idiomatic texts using **GPT-2** and aligns them with images through a **CLIP-based architecture**.
- . Fused text and image embeddings with an additional **classifier layer** and a **balanced focal loss** to effectively handle class imbalance.
- . Optimised the model via **hyperparameter grid search**, achieving **88.9% accuracy** and a **macro F1 score of 87.82%**, with robust zero-shot prediction capabilities.
- . Evaluated performance using **confusion matrices** and **prediction confidence metrics** and identified further enhancements to increase data diversity and improve fusion strategies.

Warehouse Reallocation Optimisation – *Stack: Python, NumPy, Matplotlib, Plotly*

- . Developed a custom **3D bin-packing algorithm** to optimise the placement of items into trucks for efficient warehouse relocation.
- . Implemented a **greedy heuristic** to maximise space utilization within truck containers, ensuring optimal item arrangement.
- . Processed datasets containing up to **999 items**, generating structured packing plans stored in JSON format.
- . Created **interactive and summary visualisations** using **Plotly** and **Matplotlib** to analyse and validate packing efficiency.
- . Enhanced operational logistics by reducing manual planning efforts and improving spatial resource management.

Telematics Sensor Anomaly Detection & Predictive Maintenance Pipeline

Stack: Python, PySpark, XGBoost, LSTM (PyTorch), Azure Blob Storage, Docker

- . Engineered an end-to-end predictive maintenance pipeline analysing multivariate, time-series telematics data (vibration, oil pressure, exhaust temperature) to forecast Remaining Useful Life (RUL) of heavy-duty powertrain components.
- . Developed a hybrid deep learning architecture combining a Convolutional Neural Network (CNN) for spatial feature extraction and an LSTM network to capture temporal degradation patterns over **10,000+ operational cycles**.
- . Optimised data preprocessing workflows using PySpark, handling missing sensor readouts and applying rolling window aggregations to clean and format a **50GB+ raw sensor dataset**, reducing processing time by **35%**.
- . Achieved an RMSE of **12.4** on the test set for RUL prediction and a classification accuracy of **91%** for imminent failure detection within a 30-cycle warning window.
- . Formulated a cost-matrix optimisation strategy to tune the model's decision threshold, prioritizing the reduction of False Negatives (unplanned breakdowns) while balancing False Positives (premature maintenance), projecting a simulated reduction in maintenance costs by **22%**.
- . Containerised the inference engine via Docker and simulated a deployment pipeline using Azure DevOps, enabling scalable, real-time batch scoring of incoming sensor streams.